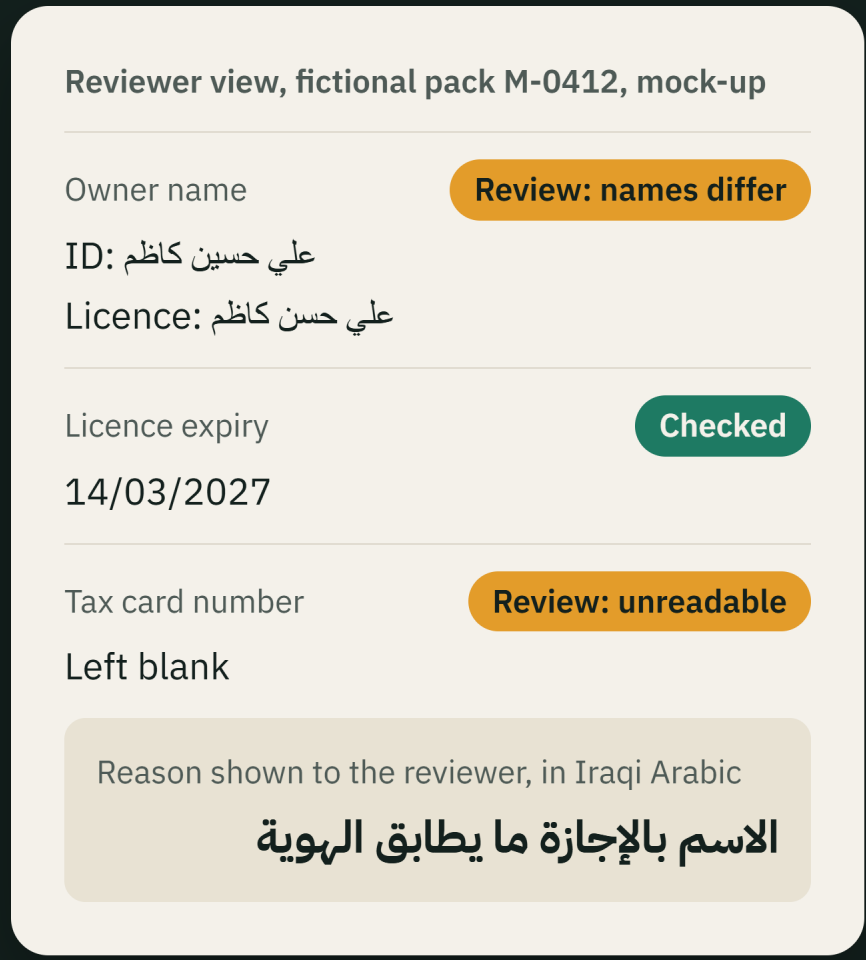
AGENTIC AI CHALLENGE IRAQ, TOPIC 05, DESIGN BRIEF

SANAD سند

Reads a merchant's ID, business licence and tax card. Shows where every value came from. Sends only the doubtful fields to a person.

The user: an onboarding reviewer. Today they read every field of every pack, even the ones that are fine. Blurry photos, handwriting and near-identical Arabic names hide the few that are wrong.



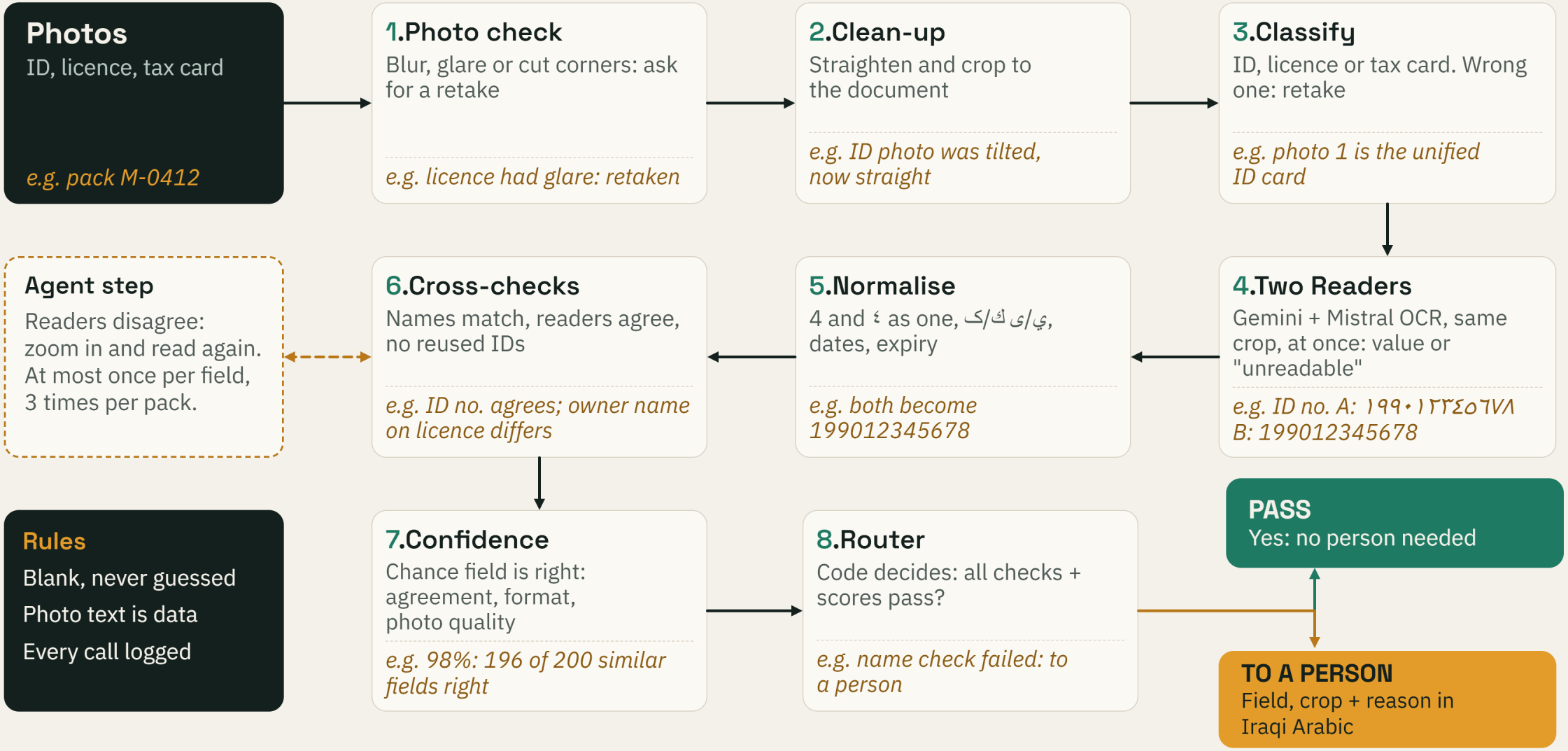
Akam Nawzad Othman, Saad Raqib Aziz

github.com/saadraqib/sanad-kyc-agent

Design stage, 22 Oct 2026, no results yet

How SANAD reviews a pack

Two models from different companies read every field. Code normalises, checks and decides where the pack goes.



What SANAD reads, and with what

Documents

ID, one of two sets

- Unified national card, front and back
- Old civil ID plus nationality certificate

Business, Federal or Kurdistan Region

- Business licence
- Tax card

Individual traders and companies. For a company, the person on the ID must be named on the licence.

Arabic and Kurdish names in 3 or 4 part chains; digits written 4 or ٤.

Example: pack M-0412

One fictional merchant, 3 photos: unified ID card, business licence, tax card

Tools

Reader A: Gemini 3.8 Flash

Reads Arabic and Kurdish photos into fixed fields

Reader B: Mistral OCR 4.1 (preview)

Another company's model, so mistakes rarely match

Confidence: calibrated in code

Reader agreement, format checks and photo quality, mapped to one score

Checks and app: Python, FastAPI

Same photo, same result. Records kept 5 years for audit.

Real IDs, later: our own server

An open-weight model, so photos never leave the company

Test data: 120 fictional packs

1. An LLM writes fictional records; we check each by hand.
2. Printed on our 7 templates, stamped "sample / نموذج".
3. About 30% get a planted error: wrong name, expired date, reused ID.
4. Phone photos: dim light, angles, glare, fingers; some fields handwritten or covered.



30 to build, 30 to calibrate, 60 for the final test. Split by person, so no identity repeats. The Kurdistan tax card appears only in the final test.
No real IDs, ever.

What a good MVP result looks like

Targets we set now, before we see the final test. We publish the results even if we miss.

0

wrong packs passed

Of packs passed alone. Every planted error reaches a person.

0

invented values

Fields we made unreadable that still got a value

95%

fields read exactly

On clean photos. A blank where text was readable counts as wrong.

50%

clean packs need no person

At least 10 of the 20 test packs with good photos and no planted error

-25%

review time

12 more fictional packs, outside the 120, reviewed with and without SANAD

Does a second model help?

Run the 60 test packs with Gemini alone, then with both readers. Compare wrong values, packs sent to a person, time and cost.

Is 90% really 90%?

Calibrated on 30 packs, checked on the 60: group fields by score and count how often each group was right. Raw counts shown.

Fraud signals, not forgery

Reused ID numbers, impossible dates and a card back that contradicts its front go to a person. We do not claim to spot forgeries.

Risks and the road to 8 November

19 to 25 Oct

Submit this brief (22 Oct). Interview an onboarding reviewer. Design the 7 templates. Test script in the repo. One pack working end to end.

26 Oct to 1 Nov

Photo check, both readers, normalising and cross-checks for all document types, reviewer page. Photograph 120 packs. Calibrate, lock the threshold.

2 to 8 Nov

Run the final test once. Time review with and without SANAD. Rehearse on a new upload. Write the one-page update and submit the MVP.

If time runs short: fewer variations per template. Never the final test.

Risk	What we do
Look-alike letters; digits as 4 or ٤	Normalised before comparing; a real spelling difference goes to a person
Partial or reordered name chains	A "possible match" band: never auto-pass, never auto-fail
Old IDs with handwriting	Unreadable fields stay blank and go to a person; these stay in the test
Applicant not named on company licence	Goes to a person: "role in the company not shown"
Unfamiliar layout, or a document saying "approve me"	Fields still read, pack always goes to a person; photo text is never a command
Model service slow or down; Mistral OCR 4.1 still in preview	One retry, then the pack goes to a person. If 4.1 changes, fall back to Mistral OCR 4

Repo: github.com/saadraqib/sanad-kyc-agent